

Hands on metabarcoding - Alpha diversity

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Alpha diversity

We will use the **vegan** package to calculate diversity. To do so, **vegan** relies on matrices, but we have our data organized in data frames to work with **dplyr** and companion packages.

We can incorporate **vegan** in the **dplyr** approach, but before doing so, we will illustrate how to calculate diversity using matrices with **vegan**.

Prepare species abundance table (1)

Start by reformatting our wide data frame to a matrix.

Select all sample columns from the **ASV_combined_clean**, excluding additional columns. You can get the samples ID from the row names of **seqtab.nochim** object.

```
# 1. select columns with samples and abundance values
ASVs_wide <- ASV_combined_clean %>%
  select(any_of(metadata_clean$Sample))
ASVs_wide$Sequence <- NULL
```

Prepare species abundance table (2)

Solution:

```
# 1. select columns with samples and abundance values  
ASVs_wide <- ASV_combined_clean[, rownames(seqtab.nochim)]
```

Prepare species abundance table (3)

The function from the **vegan** package expect species in columns and samples in rows.

Change ASVs_wide accordingly:

```
# 2. switch rows and columns with t()  
ASVs_wide_t <- ____ (ASVs_wide)
```

Prepare species abundance table (4)

Solution:

```
# 2. switch rows and columns with t()  
ASVs_wide_t <- t(ASVs_wide)
```

Calculate alpha diversity (1)

Now that we have a matrix of the species abundance table, we can directly calculate alpha diversity.

For the classic Species Richness, or ASV Richness in this context, you can use the **specnumber()** function.

Calculate Species Richness of the **ASVs_wide_t** object:

```
----(----)
```

Calculate alpha diversity (2)

Solution:

```
options(width = 60)
specnumber(ASVs_wide_t)
```

```
## FKT.006 FKT.007 FKT.008 FKT.009 FKT.010 FKT.011 FKT.012
##      616      614      554      623      607      609      560
## FKT.013 FKT.014 FKT.015 FKT.019 FKT.020 FKT.021 FKT.022
##      585      651      651      638      560      621      652
## FKT.023 FKT.024 FKT.025 FKT.026 FKT.027 FKT.028 FKT.029
##      532      521      699      659      604      625      616
## FKT.030 FKT.031 FKT.032 FKT.033
##      601      526      639      638
```


Calculate alpha diversity (3)

You can also calculate other alpha diversity indices, using the **diversity()** function from `vegan`.

Calculate:

- Shannon index;
- Simpson index;
- inverse Simpson index.

Note: use **help(diversity)** to explore the function details.

```
# Shannon index  
____(ASVs_wide_t, index = "shannon")  
# Simpson index  
____(ASVs_wide_t, index = ____)  
# inverse Simpson index  
____(____, ____)
```

Calculate alpha diversity (4)

Solution:

```
# Shannon index  
diversity(ASVs_wide_t, index = "shannon")  
# Simpson index  
diversity(ASVs_wide_t, index = "simpson")  
# inverse Simpson index  
diversity(ASVs_wide_t, index = "invsimpson")
```

Overview results with base R (1)

Using base R plots, we can have a quick look at the results.

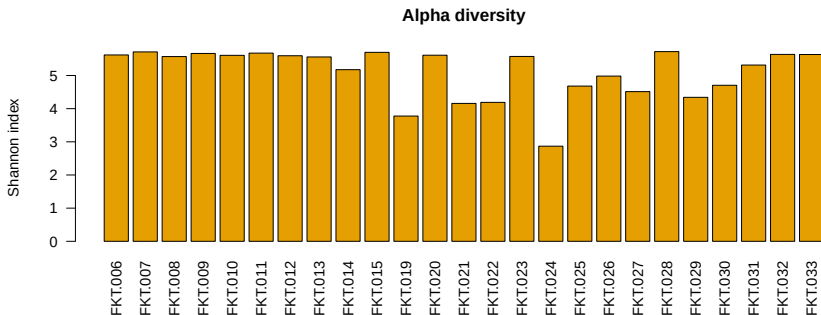
Make a bar plot with the Shannon index:

```
shannon_diversity <- diversity(ASVs_wide_t, index = "shannon")  
barplot(height = ____, las = 2, col = qualitative_colors[1],  
        main = "Alpha diversity", ylab = "Shannon index")
```

Overview results with base R (2)

Solution:

```
shannon_diversity <- diversity(ASVs_wide_t, index = "shannon")  
barplot(height = shannon_diversity,  
        las = 2, col = qualitative_colors[1],  
        main = "Alpha diversity", ylab = "Shannon index")
```



Optimize diversity calculation with dplyr grammar (1)

The previous approach relied on matrices, which makes it hard to integrate metadata and taxonomic information.

Now, we are going back to using data frames, which allow for more powerful analyses.

Optimize diversity calculation with dplyr grammar (2)

Let's start by making a summary of all diversity indices for each sample, keeping the metadata. Recall that we previously made an long table with everything integrated, **ASVs_full**.

Using the **summarise()** function from dplyr, calculate the previous diversity indices:

```
diversity_summary <- ASVs_full %>%  
  group_by(Sample, Experiment, Treatment,  
            Concentration, Replicate) %>%  
  summarise(speciesRichness = ____ (Abundance),  
            shannon = ____ (Abundance,  
                             index = "shannon"),  
            simpson = diversity(____,  
                                index = "____"),  
            invsimpson = ____ (____,  
                               index = "____"))
```

Optimize diversity calculation with dplyr grammar (3)

Solution:

```
diversity_summary <- ASVs_full %>%  
  group_by(Sample, Experiment, Treatment,  
            Concentration, Replicate) %>%  
  summarise(speciesRichness = specnumber(Abundance),  
            shannon = diversity(Abundance,  
                                index = "shannon"),  
            simpson = diversity(Abundance,  
                                index = "simpson"),  
            invsimpson = diversity(Abundance,  
                                   index = "invsimpson"))
```

```
## `summarise()` has grouped output by 'Sample', 'Experiment',  
## 'Treatment', 'Concentration'. You can override using the  
## `.groups` argument.
```

Optimize diversity calculation with dplyr grammar (4)

Inspect the new object, **diversity_summary**.

```
colnames(____)  
View(____)
```


Optimize diversity calculation with dplyr grammar (5)

Solution:

```
colnames(diversity_summary)
```

```
## [1] "Sample"          "Experiment"      "Treatment"  
## [4] "Concentration"   "Replicate"       "speciesRichness"  
## [7] "shannon"         "simpson"         "invsimpson"
```

```
View(diversity_summary)
```

Overview the summary with ggplot2 (1)

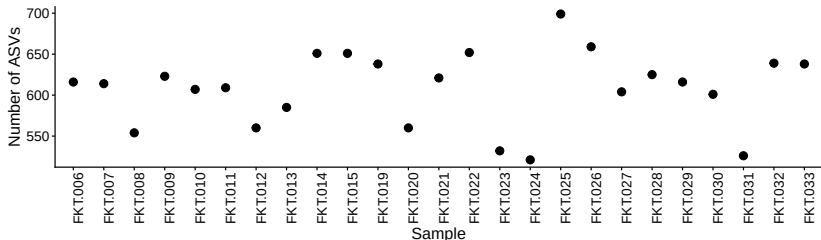
Using the **diversity_summary** data frame and ggplot2, plot species richness:

```
---- %>%  
  ggplot(aes(x = ____, y = ____)) +  
  geom_point(size = 3) +  
  theme_classic() +  
  theme(axis.text.x = element_text(size = 12, angle = 90),  
        axis.text.y = element_text(size = 12),  
        axis.title = element_text(size = 14)) +  
  labs(y = "Number of ASVs")
```

Overview the summary with ggplot2 (2)

Solution:

```
diversity_summary %>%  
  ggplot(aes(x = Sample, y = speciesRichness)) +  
  geom_point(size = 3) +  
  theme_classic() +  
  theme(axis.text.x = element_text(size = 12, angle = 90),  
        axis.text.y = element_text(size = 12),  
        axis.title = element_text(size = 14)) +  
  labs(y = "Number of ASVs")
```



Reflect the experiment in the plot (1)

Plot **Species Richness** as a function of **Concentration**, coloring points by **Replicate**, and divide the **Experiments** into two grids.

```
diversity_summary %>%  
  ggplot(aes(x = _____, y = _____,  
             col = _____)) +  
  geom_point(size = 3) +  
  geom_line(aes(group = Replicate)) +  
  facet_grid(~_____) +  
  scale_x_log10() +  
  theme_classic() +  
  scale_color_manual(values = qualitative_colors) +  
  labs(y = "Species richness",  
       x = "Concentration (M) in Log10 scale")
```

Reflect the experiment in the plot (2)

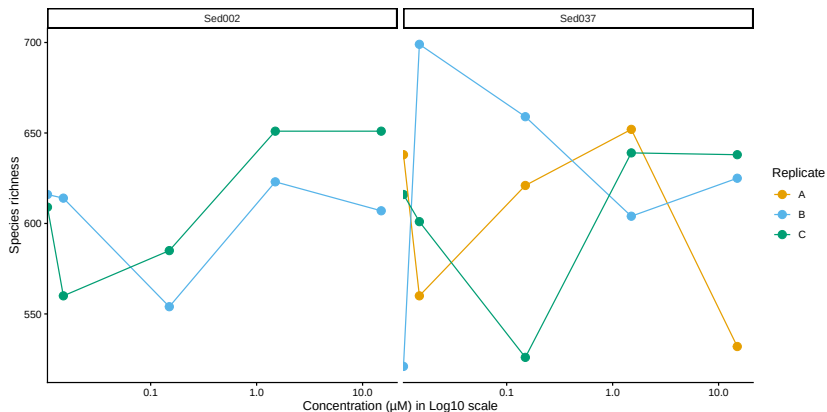
Solution:

```
diversity_summary %>%  
  ggplot(aes(x = Concentration, y = speciesRichness,  
             col = Replicate)) +  
  geom_point(size = 3) +  
  geom_line(aes(group = Replicate)) +  
  facet_grid(~Experiment) +  
  scale_x_log10() +  
  theme_classic() +  
  scale_color_manual(values = qualitative_colors) +  
  labs(y = "Species richness",  
       x = "Concentration (µM) in Log10 scale")
```

See the plot in the next slide.

Reflect the experiment in the plot (3)

```
## Warning in scale_x_log10(): log-10 transformation introduced  
## log-10 transformation introduced infinite values.
```



Reflect the experiment in the plot (4)

Repeat the same plot with factor variable, **Treatment**, instead of **Concentration**.

```
diversity_summary %>%  
  ggplot(aes(x = ____, y = speciesRichness,  
             col = Replicate)) +  
  geom_point(size = 6) +  
  geom_line(aes(group = Replicate), lwd = 2) +  
  facet_grid(~Experiment) +  
  theme_classic() +  
  scale_color_manual(values = qualitative_colors) +  
  labs(y = "Species richness",  
       x = "Concentration (\U00B5M)")
```

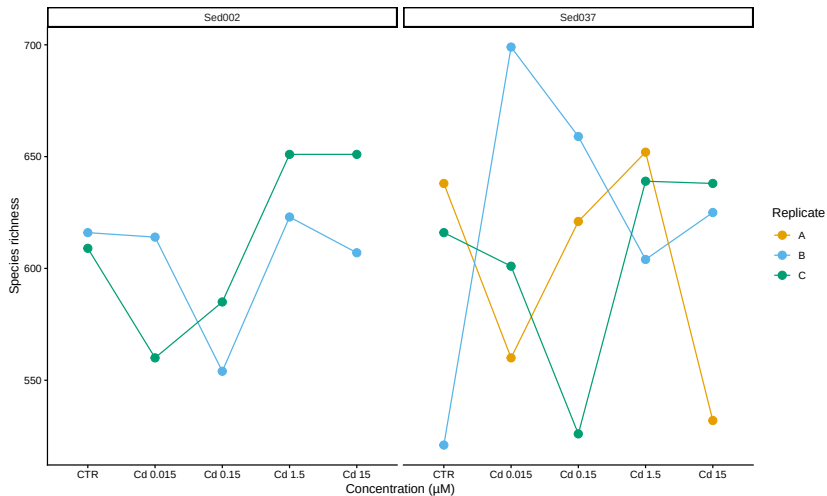
Reflect the experiment in the plot (5)

Solution:

```
diversity_summary %>%  
  ggplot(aes(x = Treatment, y = speciesRichness,  
             col = Replicate)) +  
  geom_point(size = 3) +  
  geom_line(aes(group = Replicate)) +  
  facet_grid(~Experiment) +  
  theme_classic() +  
  scale_color_manual(values = qualitative_colors) +  
  labs(y = "Species richness", x = "Concentration (µM)")
```

See the plot in the next slide.

Reflect the experiment in the plot (6)



Perfecting your plot (1)

Store the previous ggplot in an object, **plot_1**, and increase the size of the points to **6** and the width of the lines to **2**.

```
plot_1 <- diversity_summary %>%  
  ggplot(aes(x = Treatment, y = speciesRichness,  
             col = Replicate)) +  
  geom_point(size = 6) +  
  geom_line(aes(group = Replicate), lwd = 2) +  
  facet_grid(~Experiment) +  
  theme_classic() +  
  scale_color_manual(values = qualitative_colors) +  
  labs(y = "Species richness",  
       x = "Concentration (µM)")
```

Perfecting your plot (2)

Solution:

```
plot_1 <- diversity_summary %>%  
  ggplot(aes(x = Treatment, y = speciesRichness,  
             col = Replicate)) +  
  geom_point(size = 6) +  
  geom_line(aes(group = Replicate), lwd = 2) +  
  facet_grid(~Experiment) +  
  theme_classic() +  
  scale_color_manual(values = qualitative_colors) +  
  labs(y = "Species richness",  
       x = "Concentration (\U00B5M)")
```

Perfecting your plot (3)

Use the **theme** function to add editing details, on top of **plot_1**.

```
---- +
---- (axis.title = element_text(size = 16),
      axis.text.x = element_text(size = 14,
                                  angle = 45, hjust = 1),
      axis.text.y = element_text(size = 14),
      strip.text = element_text(size = 16, face = "bold"),
      strip.background = element_blank(),
      legend.title = element_text(size = 16),
      legend.text = element_text(size = 16),
      legend.position = "top")
```

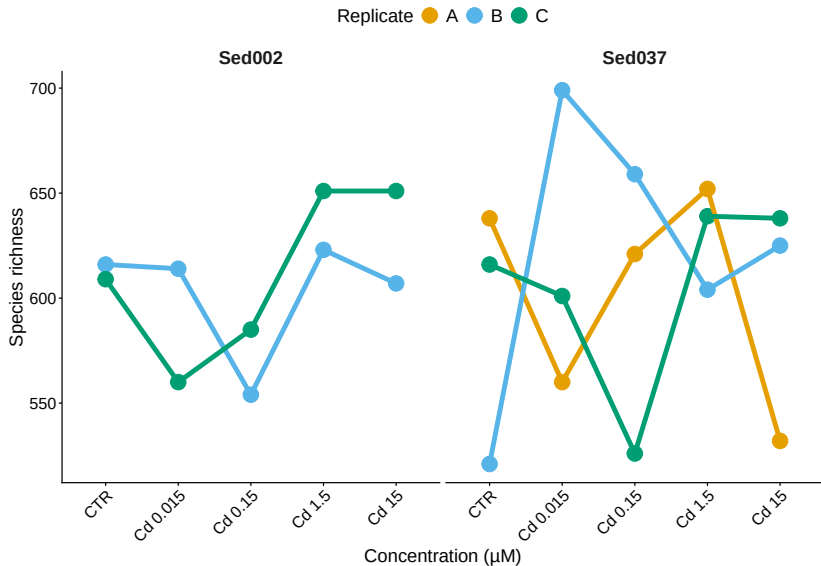
Perfecting your plot (4)

Solution:

```
plot_1 +  
  theme(axis.title = element_text(size = 16),  
        axis.text.x = element_text(size = 14,  
                                     angle = 45, hjust = 1),  
        axis.text.y = element_text(size = 14),  
        strip.text = element_text(size = 16, face = "bold"),  
        strip.background = element_blank(),  
        legend.title = element_text(size = 16),  
        legend.text = element_text(size = 16),  
        legend.position = "top")
```

The plot is in the next slide.

Perfecting your plot (5)



Repeat the same analysis for Shannon index (1)

We can repeat the previous steps for the Shannon index and see what different information each index provides.

Make a ggplot object, **plot_shannon**:

```
---- <- diversity_summary %>%  
  ggplot(aes(x = Treatment, y = ----,  
             col = Replicate)) +  
  geom_point(size = 6) +  
  geom_line(aes(group = Replicate), lwd = 2) +  
  facet_grid(~Experiment) +  
  theme_classic() +  
  scale_color_manual(values = qualitative_colors) +  
  labs(y = "Shannon",  
       x = "Concentration (\U00B5M)")
```

Repeat the same analysis for Shannon index (2)

Solution:

```
plot_shannon <- diversity_summary %>%  
  ggplot(aes(x = Treatment, y = shannon,  
             col = Replicate)) +  
  geom_point(size = 6) +  
  geom_line(aes(group = Replicate), lwd = 2) +  
  facet_grid(~Experiment) +  
  theme_classic() +  
  scale_color_manual(values = qualitative_colors) +  
  labs(y = "Shannon",  
       x = "Concentration (µM)")
```


Repeat the same analysis for Shannon index (3)

Improve **plot_shannon** with the **theme()** function:

```
----- +
    ____ (axis.title = element_text(size = 16),
          axis.text.x = element_text(size = 14,
                                     angle = 45, hjust = 1),
          axis.text.y = element_text(size = 14),
          strip.text = element_text(size = 16, face = "bold"),
          strip.background = element_blank(),
          legend.title = element_text(size = 16),
          legend.text = element_text(size = 16),
          legend.position = "top")
```

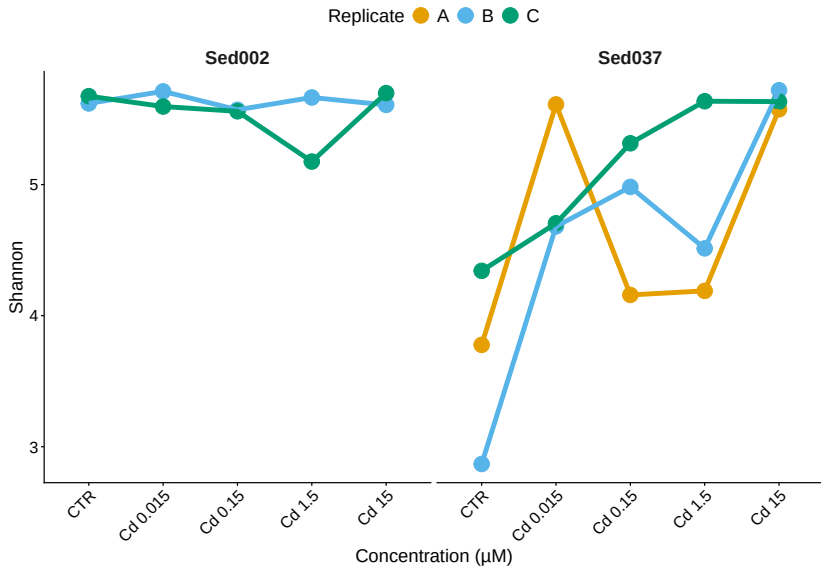
Repeat the same analysis for Shannon index (4)

Solution:

```
plot_shannon +  
  theme(axis.title = element_text(size = 16),  
        axis.text.x = element_text(size = 14,  
                                     angle = 45, hjust = 1),  
        axis.text.y = element_text(size = 14),  
        strip.text = element_text(size = 16, face = "bold"),  
        strip.background = element_blank(),  
        legend.title = element_text(size = 16),  
        legend.text = element_text(size = 16),  
        legend.position = "top")
```

See the plot in the next slide.

Repeat the same analysis for Shannon index (5)



Focus on Sed037 (1)

Compare Species richness and Shannon index for Sed037.

Repeat the previous figures, but filter for the **Sed037** Experiment.

```
Sed0037_experiment <- diversity_summary %>% filter(____)
```

Focus on Sed037 (2)

Solution:

```
Sed0037_experiment <- diversity_summary %>%  
  filter(Experiment == "Sed037")
```

Focus on Sed037 (3)

Repeat the previous plot for species richness and shannon index.

Make a plot with species richness, and store it in the object **Sed037_sr_plot**:

```
____ <- Sed0037_experiment %>%  
  ggplot(aes(x = Treatment, y = ____,  
             col = Replicate)) +  
  geom_point(size = 6) +  
  geom_line(aes(group = Replicate), lwd = 2) +  
  geom_hline(yintercept = c(550, 650), lty = "dashed") +  
  theme_classic() +  
  scale_color_manual(values = qualitative_colors) +  
  labs(y = "Species richness",  
       x = "Concentration (\U00B5M)")
```

Focus on Sed037 (4)

Solution:

```
Sed037_sr_plot <- Sed0037_experiment %>%  
  ggplot(aes(x = Treatment, y = speciesRichness,  
             col = Replicate)) +  
  geom_point(size = 3) +  
  geom_line(aes(group = Replicate), lwd = 1) +  
  geom_hline(yintercept = c(550, 650), lty = "dashed") +  
  theme_classic() +  
  scale_color_manual(values = qualitative_colors) +  
  labs(y = "Species richness",  
       x = "Concentration (\u00B5M)")
```

Focus on Sed037 (5)

Repeat the same step for Shannon index, store the plot in **Sed037_sh_plot** object.

```
Sed037_sh_plot <- Sed0037_experiment %>%  
  ggplot(aes(x = Treatment, y = shannon,  
             col = Replicate)) +  
  geom_point(size = 3) +  
  geom_line(aes(group = Replicate), lwd = 1) +  
  theme_classic() +  
  scale_color_manual(values = qualitative_colors) +  
  labs(y = "Shannon",  
       x = "Concentration (\U00B5M)")
```


Focus on Sed037 (6)

Add the previous edits to these new plots.

```
Sed037_sr_plot <- Sed037_sr_plot +  
  theme(axis.title = element_text(size = 16),  
        axis.text.x = element_text(size = 14,  
                                     angle = 45, hjust = 1),  
        axis.text.y = element_text(size = 14),  
        strip.text = element_text(size = 16, face = "bold"),  
        strip.background = element_blank(),  
        legend.title = element_text(size = 16),  
        legend.text = element_text(size = 16),  
        legend.position = "top")
```

Focus on Sed037 (7)

Cont.

```
Sed037_sh_plot <- Sed037_sh_plot +  
  theme(axis.title = element_text(size = 16),  
        axis.text.x = element_text(size = 14,  
                                     angle = 45, hjust = 1),  
        axis.text.y = element_text(size = 14),  
        strip.text = element_text(size = 16, face = "bold"),  
        strip.background = element_blank(),  
        legend.title = element_text(size = 16),  
        legend.text = element_text(size = 16),  
        legend.position = "top")
```

Focus on Sed037 (8)

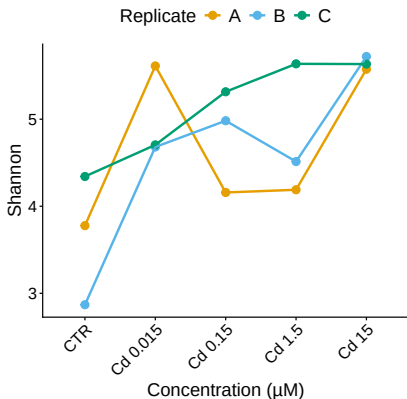
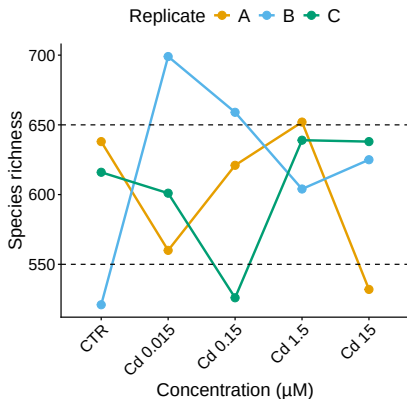
To combine two different ggplot objects, we can use the function **grid.arrange()** from the gridgridExtra package.

Combine the objects **Sed037_sr_plot** and **Sed037_sh_plot** in a single plot:

```
____(____, ____, ncol = 2, nrow = 1)
```

Focus on Sed037 (9)

Solution:



So far, what can we conclude?

Does diversity increase for different treatments? (1)

From the previous plot, it seems that:

- The number of ASVs (Species Richness) does not show any trend, with most samples within a range of 550 to 650 ASVs;
- However, the values of Shannon index seem to be increasing.

Does diversity increase for different treatments? (2)

To confirm if the trends of increasing diversity with Cadmium treatment is statistically significant, we could use the **one-way repeated measures ANOVA** test.

Why this test:

- More than three groups (ANOVA test);
- the groups are dependent of each other, *i.e.*, one treatment influences the outcome of the next treatment (repeated measures ANOVA test);
- We have one response variable, the Shannon index (one-way repeated measures ANOVA).

Does diversity increase for different treatments? (2)

Some problems with this test:

- Our example has **low sample size**;
- Many requirements (parametric test);
 - normal distribution;
 - no extreme outliers ;
 - Sphericity (same variation between groups).

Verify parametric requisits - extreme outliers (1)

Check if there are extreme outliers in the Shannon index scores of the Sed037 experiment, using the **identify_outliers()** function from **rstatix**.

```
# identify extreme outliers
Sed0037_experiment %>%
  group_by(Treatment) %>%
  ____("shannon")
```


Verify parametric requisits - extreme outliers (2)

Solution:

```
# identify extreme outliers
Sed0037_experiment %>%
  group_by(Treatment) %>%
  identify_outliers("shannon")
```

```
## [1] Treatment      Sample           Experiment
## [4] Concentration  Replicate        speciesRichness
## [7] shannon         simpson          invsimpson
## [10] is.outlier       is.extreme
## <0 rows> (or 0-length row.names)
```

There are no outliers

Verify parametric requisits - normal distribution (1)

To verify the normal distribution, we can use the **Shapiro-Wilk test**, with the **rstatix** package.

The null hypothesis is that the data follows a normal distribution. Thus, if the test is **significant** ($p < 0.05$), the distribution is **non-normal**.

Our objective is to get $p > 0.05$ (accept the null hypothesis that the data follows the normal distribution).

Use the **shapiro_test()** function from **rstatix**:

```
Sed0037_experiment %>%  
  group_by(Treatment) %>%  
  ____(shannon)
```

Verify parametric requisits - normal distribution (2)

Solution:

```
Sed0037_experiment %>%  
  group_by(Treatment) %>%  
  shapiro_test(shannon)
```

```
## # A tibble: 5 x 4  
##   Treatment variable statistic      p  
##   <fct>      <chr>      <dbl> <dbl>  
## 1 CTR        shannon      0.982 0.744  
## 2 Cd 0.015   shannon      0.769 0.0434  
## 3 Cd 0.15    shannon      0.943 0.541  
## 4 Cd 1.5     shannon      0.908 0.411  
## 5 Cd 15      shannon      0.988 0.789
```

One single treatment (Cd 0.015) does not follow the normal distribution, but we are going to move forward with it.

Verify parametric requisits - sphericity

Sphericity will be integrated in the ANOVA test itself, we don't need to test before the test in this context.

One-way repeated measures ANOVA (1)

To calculate one-way repeated measures ANOVA, use the function **anova_test()**, from **rstatix** package, together with **dplyr**. Store the test result in a new object, called **shannon_anova** and get a table with the result, using **get_anova_table()** function.

```
____ <- Sed0037_experiment %>%  
  ungroup() %>%  
  ____ (dv = shannon,  
        wid = Replicate,  
        within = Treatment)  
  
#  
get_anova_table(____)
```

One-way repeated measures ANOVA (2)

Solution:

```
shannon_anova <- Sed0037_experiment %>%  
  ungroup() %>%  
  anova_test(dv = shannon,  
             wid = Replicate,  
             within = Treatment)  
  
#  
get_anova_table(shannon_anova)
```

```
## ANOVA Table (type III tests)
```

```
##
```

##	Effect	DFn	DFd	F	p	p<.05	ges
## 1	Treatment	4	8	4.686	0.03	*	0.634

One-way repeated measures ANOVA (3)

```
## ANOVA Table (type III tests)
##
##      Effect DFn DFd      F      p p<.05      ges
## 1 Treatment   4   8 4.686 0.03      * 0.634
```

Test information:

- DFn/DFd - Degrees of freedom from the numerator/denominator;
- F - F statistic (value that is compared with F distribution);
- ges - generalized effect size (within variability, from 0 to 1).

Result: The mean values of Shannon index are statistically significantly different between treatments ($p < 0.05$). However, **they were significant by very little.**

What treatments were responsible for the significant differences?

Post-hoc test - pairwise paired t-test (1)

Compare all combinations of treatments with paired t-test to identify which combinations were statistically significantly different.

Use the function **pairwise_t_test()** from rstatix package, and apply **bonferroni** method to correct for Type I errors:

```
Sed0037_experiment %>%  
  ungroup() %>%  
  ----(shannon ~ Treatment, paired = TRUE,  
        p.adjust.method = ----)
```


Post-hoc test - pairwise paired t-test (2)

Solution:

```
options(width = 50)
ptt <- Sed0037_experiment %>%
  ungroup() %>%
  pairwise_t_test(shannon ~ Treatment, paired = TRUE,
                  p.adjust.method = "bonferroni")
head(ptt, n = 3)
```

```
## # A tibble: 3 x 10
##   .y.      group1 group2      n1      n2 statistic      df
##   <chr> <chr>   <chr>   <int> <int>     <dbl> <dbl>
## 1 shann~ CTR    Cd 0.~      3      3     -2.75      2
## 2 shann~ CTR    Cd 0.~      3      3     -2.27      2
## 3 shann~ CTR    Cd 1.5      3      3     -3.05      2
## # i 3 more variables: p <dbl>, p.adj <dbl>,
## #   p.adj.signif <chr>
```

Post-hoc test - pairwise paired t-test (3)

Verify if any pairwise test was significant:

```
ptt %>%  
  filter(p.adj.signif ---- "ns")
```

Post-hoc test - pairwise paired t-test (4)

Solution:

```
ptt %>%  
  filter(p.adj.signif != "ns")
```

```
## # A tibble: 0 x 10  
## # i 10 variables: .y. <chr>, group1 <chr>,  
## #   group2 <chr>, n1 <int>, n2 <int>,  
## #   statistic <dbl>, df <dbl>, p <dbl>,  
## #   p.adj <dbl>, p.adj.signif <chr>
```

All pairwise tests were **not significant**. Recall that one treatment was not normally distributed and that the ANOVA test was significant by very little.

We cannot conclude with confidence that the Shannon index increased significantly with different treatments.